Math 661 Applied Statistics

Module 2 Assignment Solutions

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Task 1: Gosset & Guinness - IPS Exercises 1.37, 1.38, 1.39

Context: William Sealy Gosset examined double stout sales ratios (1925 as % of 1913) for various UK regions after WWI.

Exercise 1.37: Measures of Center

(a) Mean: x = (1/12) x 1475 = **122.92**
(b) Median: Sorted: 24, 46, 66, 68, 78, 94, 111, 112, 118, 140, 190, 428
Median = (94 + 111)/2 = **102.5**
(c) Preferred: Median (102.5) is preferred due to the extreme London outlier (428).

Exercise 1.38: Measures of Spread

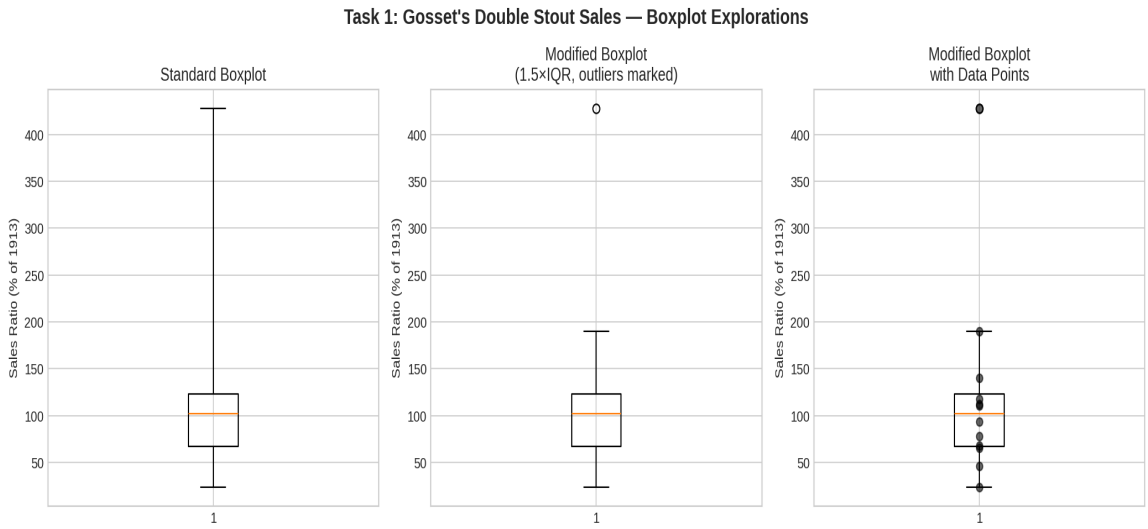
(a) Standard Deviation: s = **105.73**
(b) Quartiles: Q1 = 67.0, Q3 = 129.0
Five-Number Summary: Min=24, Q1=67, Median=102.5, Q3=129, Max=428
(c) Preferred: IQR (62.0) is preferred because it's resistant to the London outlier.

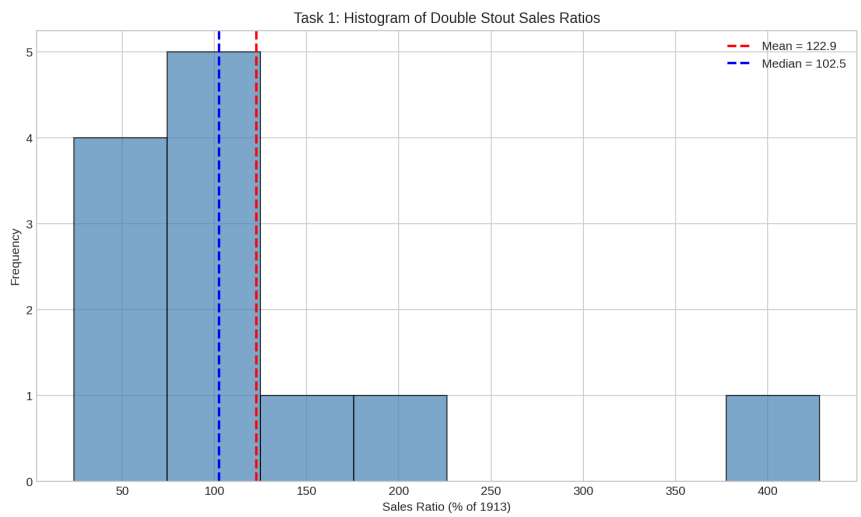
Exercise 1.39: Outlier Detection

(a) IQR: 129.0 - 67.0 = **62.0**
(b) Fences: Lower = 67 - 1.5(62) = -26.0, Upper = 129 + 1.5(62) = 222.0
Outlier: London (428) exceeds upper fence.

Stemplot:

```
 2 | 4          Stem unit = 10
 4 | 6          Leaf unit = 1
 6 | 6 8
 7 | 8          Key: 42|8 = 428
 9 | 4
11 | 1 2 8
14 | 0
19 | 0
42 | 8
```





Conclusion: London's 428% ratio is an extreme outlier. Most regions had ratios around 100%, indicating stable or declining sales. The median better represents typical post-war sales.

Task 2: Potassium from Potatoes

Context: Potassium absorption (mg) from 27 adults on a controlled diet with 40 mEq potassium from potatoes for 5 days.

(a) Stemplot:

```
Stem | Leaves        Key: 26|6 = 2660 mg
  26 | 6             Stem unit = 1000 mg
  27 | 8             Leaf unit = 100 mg
  29 | 5 5 7 7
  30 | 2 4 6 7 9
  31 | 1 1 3 4
  32 | 2 3 5 6 7 8
  34 | 8
  35 | 0 3 8
  36 | 0
  42 | 1
```

(b) Distribution Pattern:

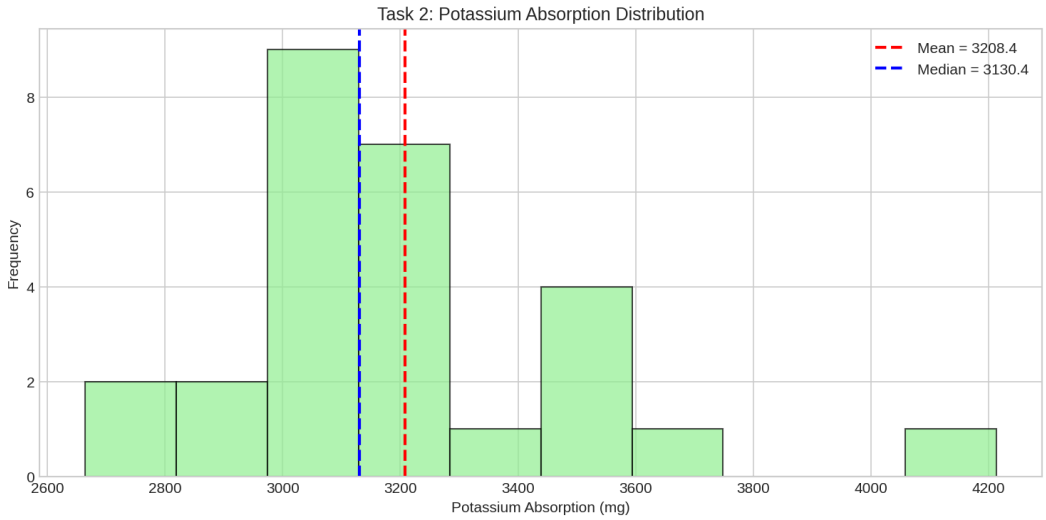
The distribution is roughly **unimodal** with observations concentrated between 2950-3300 mg. It appears **slightly right-skewed** with a tail extending toward higher values (3500-4200 mg).

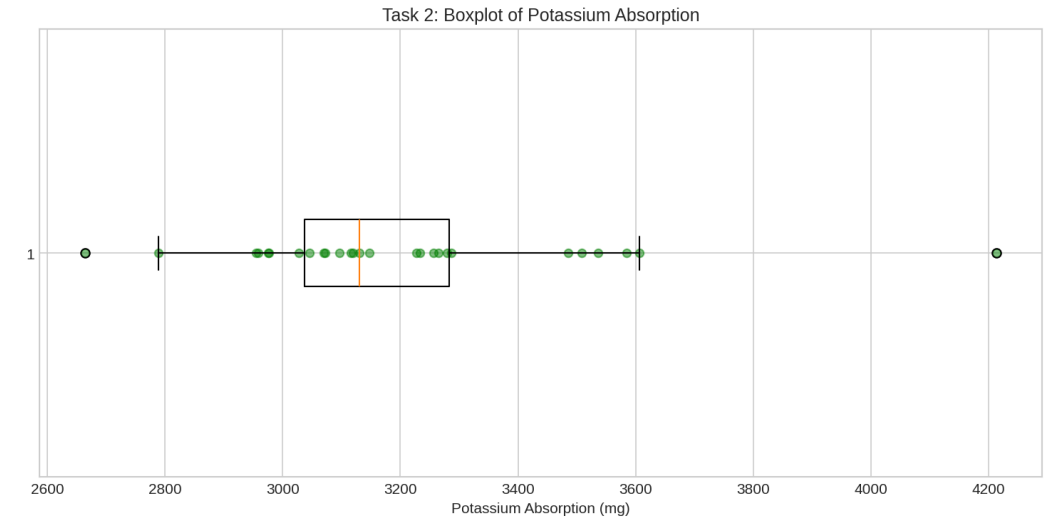
(c) Outlier Analysis:

Q1 = 3036.62, Q3 = 3283.06, IQR = 246.44
Lower fence = 3036.62 - 1.5(246.44) = **2666.95**
Upper fence = 3283.06 + 1.5(246.44) = **3652.73**
Outliers: 2664.38 mg (below), 4213.49 mg (above)

(d) Shape, Center, and Spread:

Shape: Slightly right-skewed, unimodal (Mean 3208.44 > Median 3130.37)
Center: Mean = 3208.44 mg, Median = 3130.37 mg
Spread: SD = 306.68 mg, IQR = 246.44 mg, Range = 1549.11 mg
Five-Number Summary: 2664.38, 3036.62, 3130.37, 3283.06, 4213.49


Task 2: Potassium Absorption Distribution
Mean = 3208.4
Median = 3130.4
Frequency
0
2
4
6
8
2600
2800
3000
3200
3400
3600
3800
4000
4200
Potassium Absorption (mg)


Task 2: Boxplot of Potassium Absorption
1
2600
2800
3000
3200
3400
3600
3800
4000
4200
Potassium Absorption (mg)

Task 3: Trimmed Mean - IPS Exercise 1.59

A **trimmed mean** removes a fixed percentage from each end of sorted data before computing the mean, providing resistance to outliers.

Dataset	Mean	Median	5% Trimmed	10% Trimmed
Double Stout	122.92	102.5	122.92	102.30
Potassium	3208.44	3130.37	3190.00	3189.30

Interpretation: For the stout data, the 10% trimmed mean (102.30) is close to the median, confirming the London outlier inflates the regular mean. For potassium, trimmed means are close to both mean and median, indicating fewer influential extreme values.

Task 4: Deviations from the Mean Sum to Zero

Theorem: For observations x1, x2, ..., xn with mean x-bar: Sum(xi - x-bar) = 0

Proof:
By definition: x-bar = (1/n) x Sum(xi), so n x x-bar = Sum(xi)
Sum(xi - x-bar) = Sum(xi) - Sum(x-bar) = Sum(xi) - n x x-bar = Sum(xi) - Sum(xi) = **0**

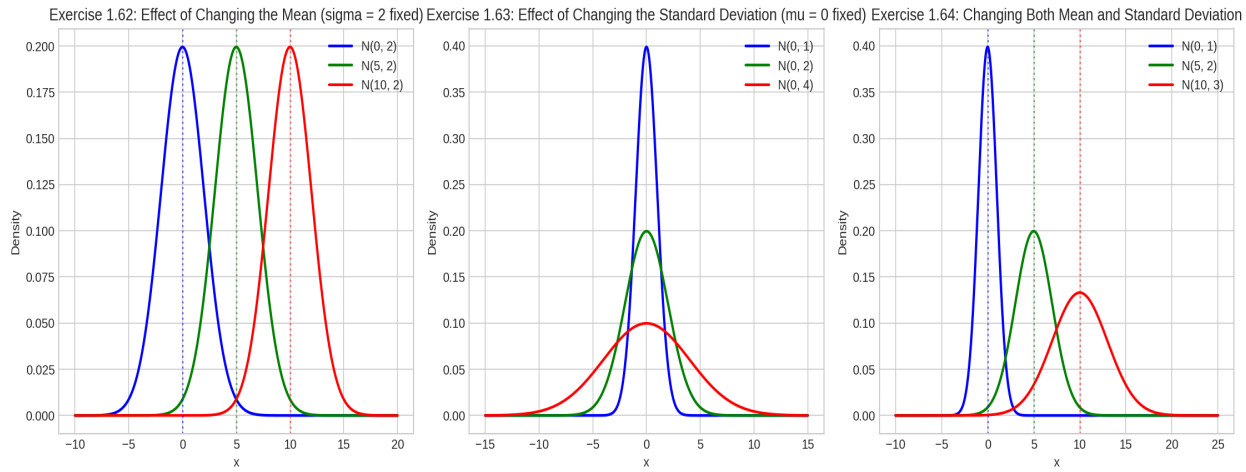
Significance: This is why variance uses **squared** deviations - raw deviations always cancel out and cannot measure spread.

Task 5: Sketching Density Curves - IPS Exercises 1.62, 1.63, 1.64

Exercise 1.62 - Effect of Changing Mean (sigma fixed):
When mu increases while sigma stays fixed, the curve **shifts right**. Shape and spread remain identical.

Exercise 1.63 - Effect of Changing Standard Deviation (mu fixed):
When sigma increases while mu stays fixed, the curve becomes **wider and shorter**. Since total area = 1, wider curves must be shorter.

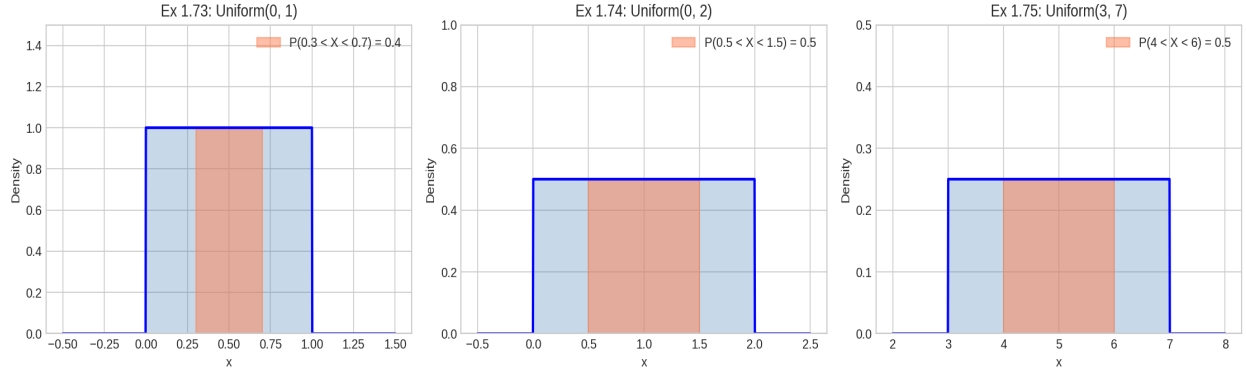
Exercise 1.64 - Changing Both:
The curve both **shifts location** and **changes spread** simultaneously.



Task 6: Uniform Distributions - IPS Exercises 1.73, 1.74, 1.75

For U(a,b): density = 1/(b-a), mean = (a+b)/2, SD = (b-a)/sqrt(12)

Exercise	Distribution	Mean	SD	Probability
1.73	U(0,1)	0.5	0.2887	P(0.3<X<0.7) = 0.40
1.74	U(0,2)	1.0	0.5774	P(0.5<X<1.5) = 0.50
1.75	U(3,7)	5.0	1.1547	P(4<X<6) = 0.50



Task 7: Normal Distribution Calculations - IPS Exercises 1.79, 1.80, 1.81

Exercise 1.79: Heights ~ N(64.5, 2.5) inches

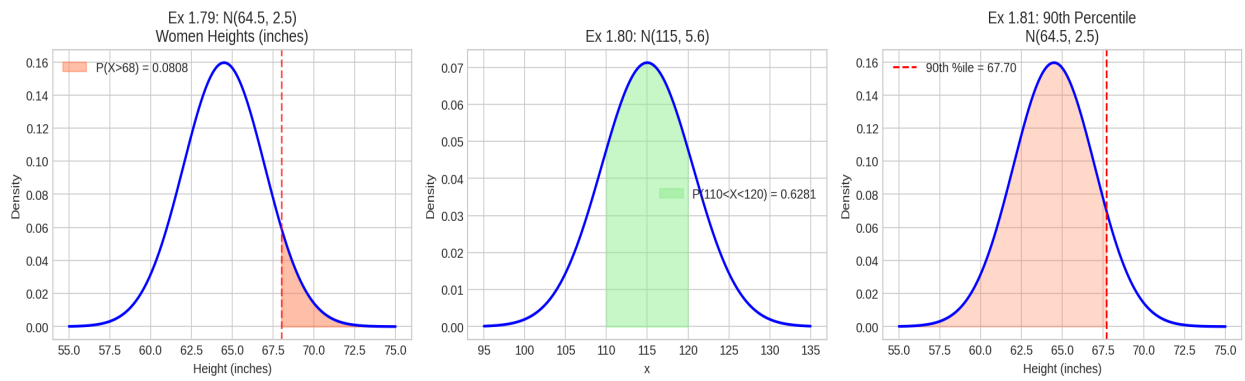
(a) P(X > 68): z = 1.40, P = **0.0808** (8.08%)
(b) P(X < 60): z = -1.80, P = **0.0359** (3.59%)
(c) P(62 < X < 67): z values = -1.00, 1.00, P = **0.6827** (68.27%)

Exercise 1.80: X ~ N(115, 5.6)

P(110 < X < 120): z values = -0.893, 0.893, P = **0.6281** (62.81%)
P(X > 125): z = 1.786, P = **0.0371** (3.71%)

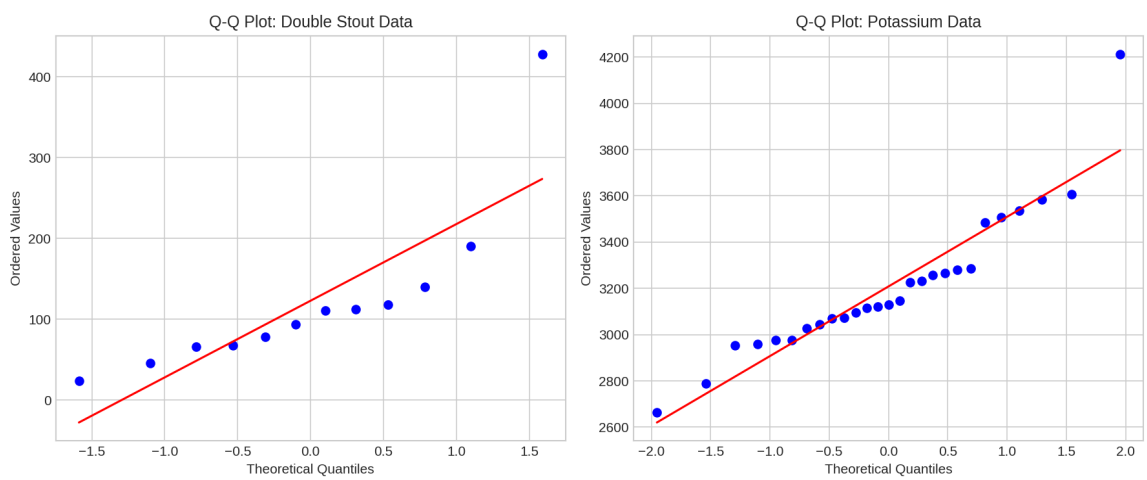
Exercise 1.81: Finding Percentiles for N(64.5, 2.5)

90th percentile: **67.70** inches (z = 1.2816)
10th percentile: **61.30** inches (z = -1.2816)
75th percentile: **66.19** inches (z = 0.6745)



Task 8: Normal Quantile Plots

Double Stout Data: Strong departure from normality. London (428) deviates dramatically, consistent with right-skewed distribution.
Potassium Data: Points generally follow the reference line with slight upper-tail deviation. Reasonably consistent with normal distribution.



Task 9: Do Identical Summary Statistics Mean Identical Distributions?

Q1: Same mean and SD - same shape?
NO. A bell-shaped and bimodal distribution can share identical mean and SD. These measures capture center and spread, not shape (symmetry, modality, tail behavior). Example: Anscombe's Quartet.

Q2: Same five-number summary - same shape?
NO. The five-number summary only specifies five values, not how observations are distributed within each quarter. Two datasets could be uniform vs. clustered within quarters with identical summaries.